Master's degree in Computer Engineering for Robotics and Smart Industry

Advanced control systems

Course assignments

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Contents

1 Introduction

This technical report outlines the coursework for the **Advanced Control Systems** class.
The code is available at Github.

2 Robot structure and kinematics

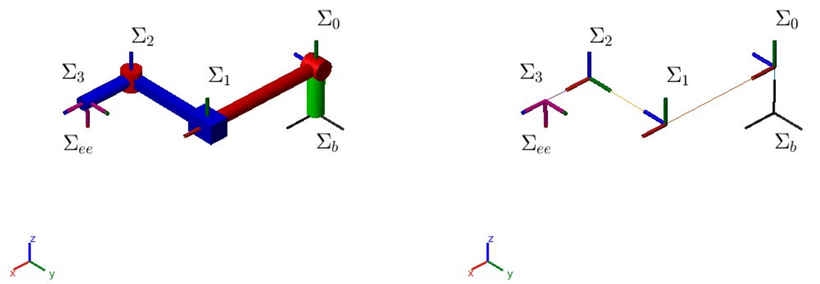


Figure 1: Robot visualization

$$\delta_{b-0} = 0.15\,\mathrm{m} \tag{1}$$
$$\delta_{0-1} = 0.4\,\mathrm{m} \tag{2}$$
$$\delta_{1-2} = 0.3\,\mathrm{m} \tag{3}$$
$$\delta_{2-3} = 0.16\,\mathrm{m} \tag{4}$$

2.1 DH table

a	α	d	θ
0	$\frac{\pi}{2}$	δ_{b-0}	0
δ_{0-1}	0	0	q_1
0	$-\frac{\pi}{2}$	$\delta_{1-2} + q_2$	0
δ_{2-3}	0	0	q_3

To have the same alignment at the end-effector as the one in from the robotic toolbox we must rotate around the y axis (of the frame Σ_3) by $\frac{\pi}{2}$ rad .

2.2 Direct kinematics

We then procede to compute the direct kinematics by chaining all the transformations from each frame Σ_i to the next(Σ_{i+1}). Each transformation is obtained by pluggin inside the following matrix the correct line of the aforementioned DH table.

$$T_i^{i-1} = \begin{bmatrix} \cos(\theta) & -\sin(\theta)\cos(\alpha) & \sin(\theta)\sin(\alpha) & a\cos(\theta) \\ \sin(\theta) & \cos(\theta)\cos(\alpha) & -\cos(\theta)\sin(\alpha) & a\sin(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) & d \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5}$$

Then

$$T_{ee}^b = T_0^b \times T_1^0 \times T_2^1 \times T_3^2 \times T_{ee}^3 \tag{6}$$

2.3 Inverse kinematics

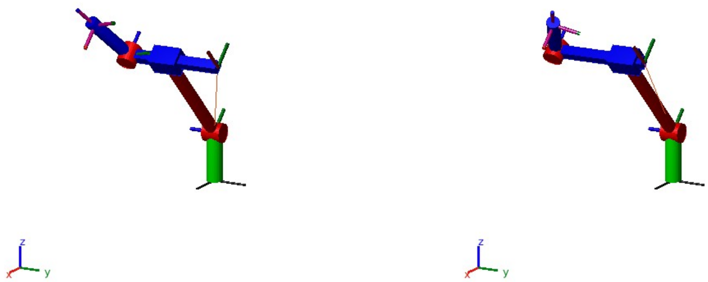
A closed form solution of the inverse kinematics problem can be found by considering only the cartesian coordinates of the end-effector (since we have only 3 DoF).

$$x_{ee} = 0.4\,\cos(q_1) + 0.16\,\cos(q_1)\cos(q_3) \tag{7}$$
$$y_{ee} = 0.16\,\sin(q_3) - q_2 - 0.3 \tag{8}$$
$$z_{ee} = 0.4\,\sin(q_1) + 0.16\,\cos(q_3)\sin(q_1) + 0.15 \tag{9}$$

By solving this system of equation(7)(8)(9) we obtain equations(10)(12)(11)

$$q_1 = \arctan\left(\frac{z_{ee} - 0.15}{x_{ee}}\right) \tag{10}$$
$$q_3 = \arccos\left(\frac{z_{ee} - 0.15 - 0.4\,\sin q_1}{0.16\,\sin(q_1)}\right) \tag{11}$$
$$q_2 = 0.16\,\sin(q_3) - 0.3 - y_{ee} \tag{12}$$

Figure(2) shows the case in which multiple valid solutions exists.



(a) Robotics system toolbox

(b) Inverse kinematics using equations (10)(11)(12)

Figure 2: Inverse kinematics solutions

2.4 Jacobian matrices

$$J(q) = \begin{bmatrix} -0.0800\sin(q_1)(2\cos(q_3)+5) & 0 & -0.1600\cos(q_1)\sin(q_3) \\ 0 & -1 & 0.1600\cos(q_3) \\ 0.0800\cos(q_1)(2\cos(q_3)+5) & 0 & -0.1600\sin(q_1)\sin(q_3) \\ 0 & 0 & -\sin(q_1) \\ -1 & 0 & 0 \\ 0 & 0 & \cos(q_1) \end{bmatrix} \quad (13)$$

The Euler angles sequence chosen for the analytical Jacobian matrix is $\boldsymbol{ZYZ}$.

3 Manipulator dynamics

3.1 Potential energy

$$\mathcal{U}(q) = g\sin(q_1) + 0.7848\cos(q_3)\sin(q_1) + 4.4145 \quad (14)$$

Where g is the gravitational acceleration.

3.2 Kinetic energy

$$\begin{aligned}\mathcal{T}(q,\dot{q}) = {} & 0.0063\,\dot{q_1}^2\,\cos(q_3)^2 + 0.0320\,\dot{q_1}^2\,\cos(q_3) + 0.2653\,\dot{q_1}^2 + \dot{q_2}^2 \\ & - 0.0800\,\dot{q_2}\,\dot{q_3}\,\cos(q_3) + 0.0128\,\dot{q_3}^2\end{aligned} \quad (15)$$

3.3 Dynamic model

3.4 Lagrangian formulation

$$B(q) = \begin{bmatrix} 0.064\,\cos(q_3) + 0.0126\,\cos(q_3)^2 + 0.5305 & 0 & 0 \\ 0 & 2 & -0.08\,cos(q_3) \\ 0 & -0.08\,cos(q_3) & 0.0256 \end{bmatrix} \tag{16}$$

$$C(q,\dot{q}) = \begin{bmatrix} -\dot{q}_3\,(0.0063\,\sin(2\,q_3)) + 0.032\,\sin(q_3) & 0 & -\dot{q}_1\,(0.0063\,\sin(2\,q_3) + 0.032\,\sin(q_3)) \\ 0 & 0 & 0.08\,\dot{q}_3\,\sin(q_3) \\ \dot{q}_1\,(0.0063\,\sin(2\,q_3) + 0.032\,\sin(q_3)) & 0 & 0 \end{bmatrix} \tag{17}$$

$$g(q) = \begin{bmatrix} 0.3924\,\cos(q_1)\,(2\,\cos(q_3) + 25) \\ 0 \\ -0.7848\,\sin(q_1)\,\sin(q_3) \end{bmatrix} \tag{18}$$

$$B(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} + g(q) = \tau \tag{19}$$

Equation(19) is the *equation of motion*, a set of 3 nonlinear second-order differential equations.

3.5 RNE formulation

Using the Newton–Euler equations it is possible to set up an algorithm composed by a **forward recursion** step in which we propagate the links velocity and acceleration from the base link to the EE and a **backward recursion** step in which we propagate the forces from the EE to the base link. Symbols[1] can be used inside the calculation using the RNE approach to obtain $B(q)$, $C(q,\dot{q}) \cdot \dot{q}$ and $g(q)$ matrices.

3.6 Dynamic model in the operational space

The aforementioned dynamic model can be transposed in the operational space via the following set of equations.

[1]Symbolic Math Toolbox

$$B_a(q) = (J_a(q)B(q)^{-1}J_a(q)^\intercal)^{-1} \tag{20}$$

$$C_a(q,\dot{q})\dot{x} = B_a(q)J_a(q)B(q)C(q,\dot{q})\dot{q} - B_a(q)\dot{J}_a(q,\dot{q})\,\dot{q} \tag{21}$$

$$g_a(q) = B_a(q)J_a(q)B(q)^{-1}g(q) \tag{22}$$

$$u_{a_e}(q) = T_a(q)^\intercal h_e \tag{23}$$

$$\tag{24}$$

3.7 Bonus: Parameters estimation

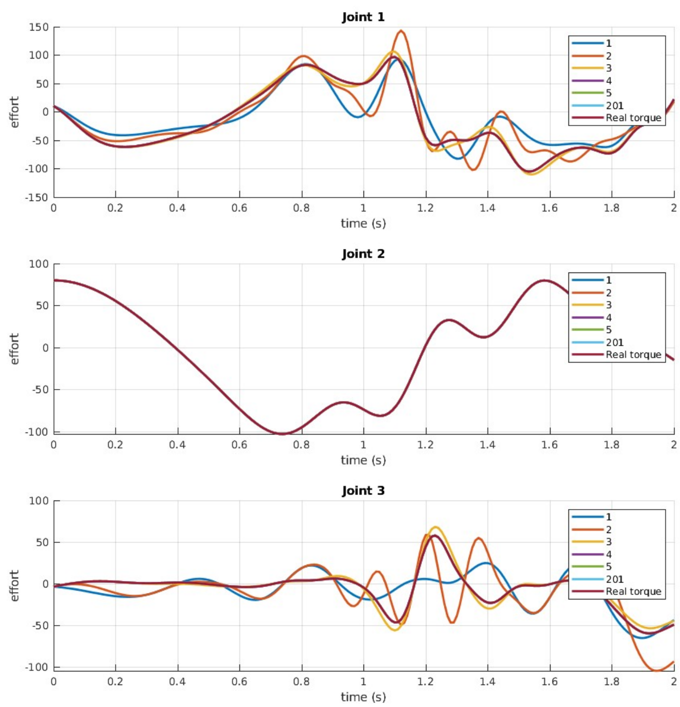


Figure 3: Parameters estimation
different simulation steps amount are taken into account

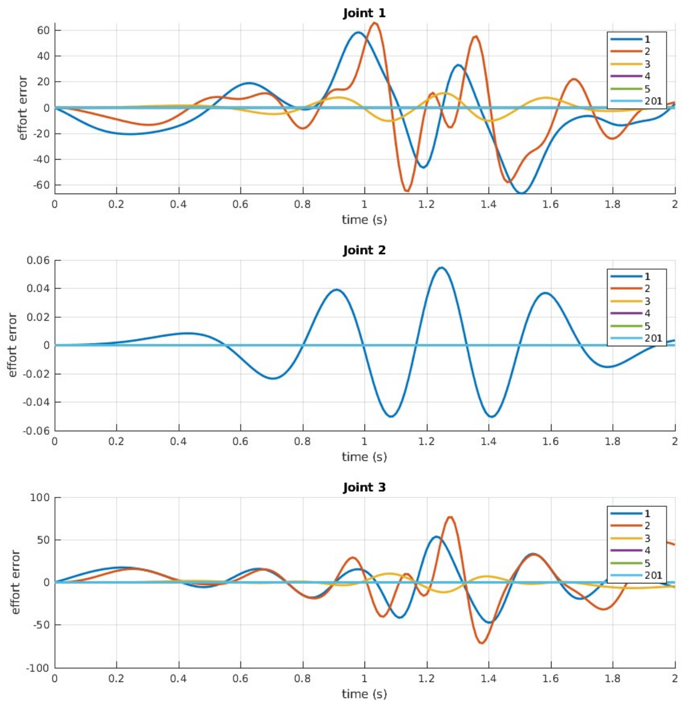


Figure 4: Parameters estimation error
different simulation steps amount are taken into account

Link # \ Property	Mass	Inertia tensor (w.r.t. CoM)
1	1	$\begin{bmatrix} 0.0002 & 0 & 0 \\ 0 & 0.0800 & 0 \\ 0 & 0 & 0.0800 \end{bmatrix}$
2	1	$\begin{bmatrix} 0.0076 & 0 & 0 \\ 0 & 0.0150 & 0 \\ 0 & 0 & 0.0076 \end{bmatrix}$
3	1	$\begin{bmatrix} 0.0002 & 0 & 0 \\ 0 & 0.0128 & 0 \\ 0 & 0 & 0.0128 \end{bmatrix}$

Table 1: Real robot parameters

Link # \ Property	Mass	Inertia tensor (w.r.t. CoM)
1	1	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0.0406 \end{bmatrix}$
2	1	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.0406 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
3	1	$\begin{bmatrix} 0.0140 & 0 & 0 \\ 0 & 0.0266 & 0 \\ 0 & 0 & 0.0128 \end{bmatrix}$

Table 2: Estimated robot parameters

4 Control schemes

4.1 Joint Space PD control law with gravity compensation

$$\tau = g(q_d) + K_P(q_d - q) - K_D\dot{q} \tag{25}$$

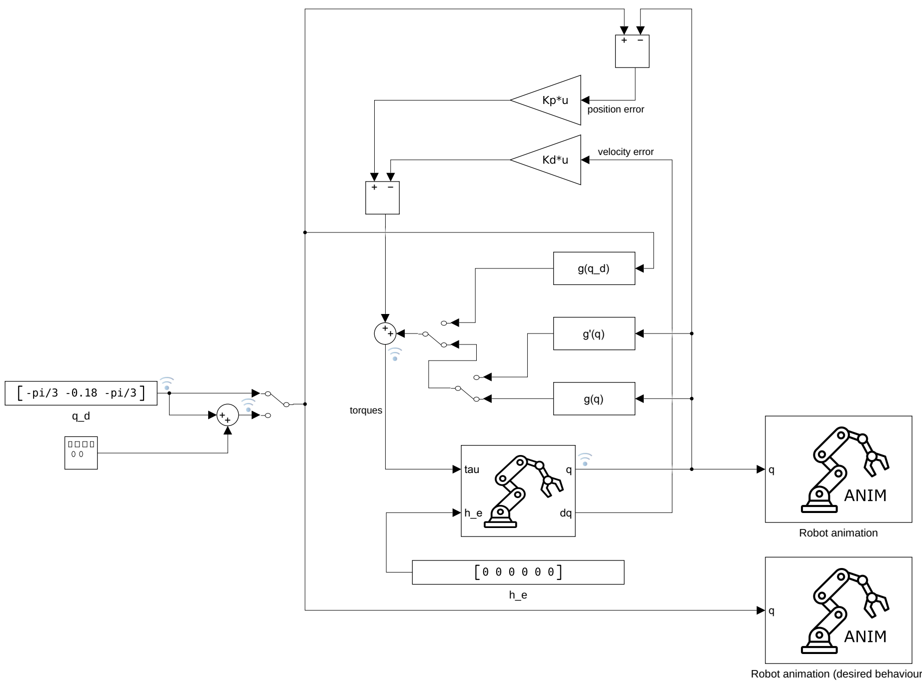
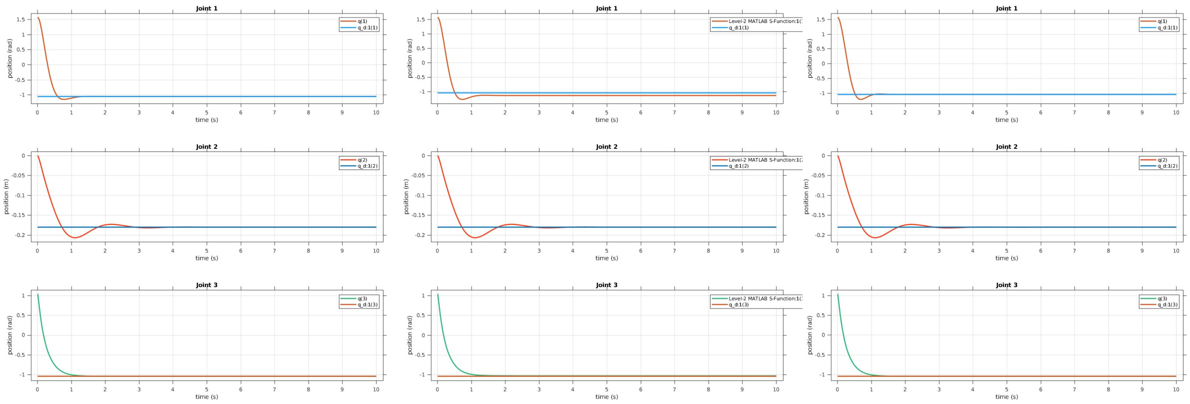


Figure 5: Joint Space PD control law with gravity compensation control scheme



(a) (b) compensated gravity term is $g'(q)$ (c) compensated gravity term is $g(q_d)$

Figure 6: Joint Space PD control law with gravity compensation

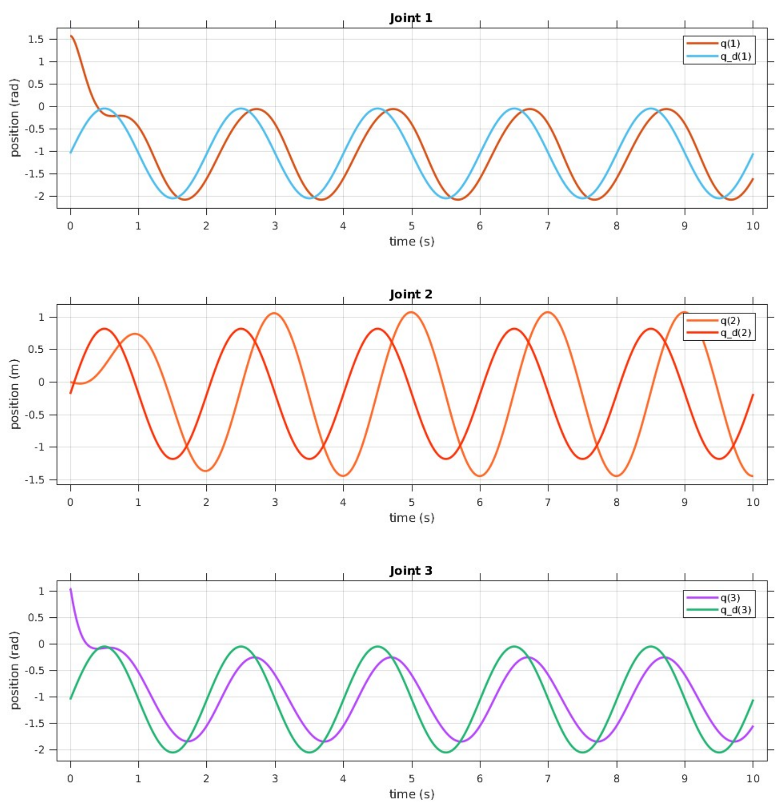


Figure 7: Joint Space PD control law with gravity compensation tracking task

4.2 Joint Space Inverse Dynamics control law

Eq(26) is the nonlinear compensation and decoupling inner loop, eq(27) is the outer control loop.

$$\tau = B(q)\ y + n(q, q\dot{q}) \tag{26}$$

$$\ddot{q} = y \tag{27}$$

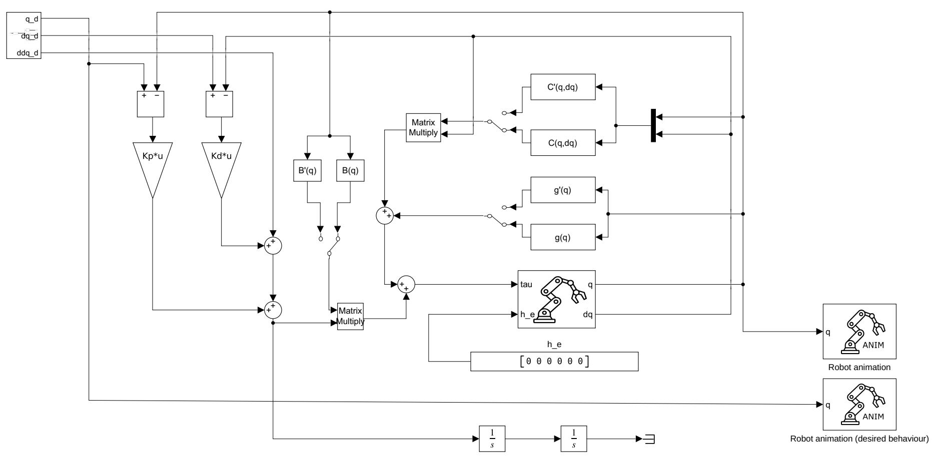
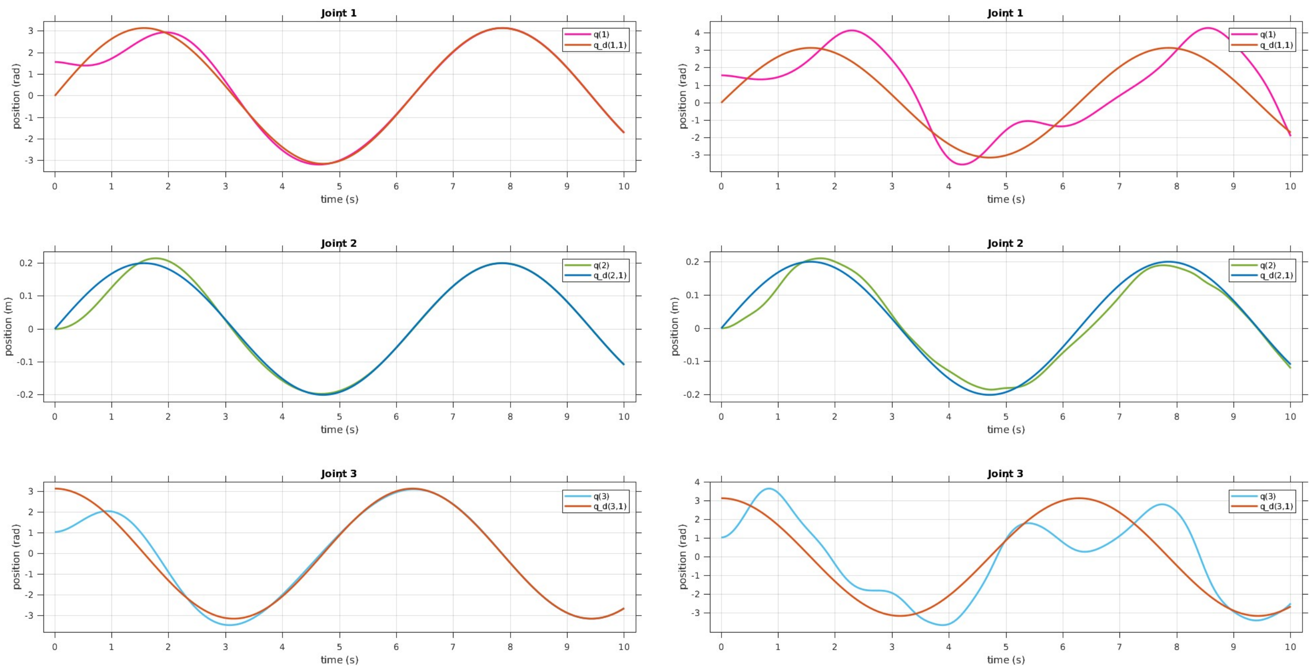


Figure 8: Joint Space Inverse Dynamics control law control scheme



(a) (b) dynamics matrices are a little off

Figure 9: Joint Space Inverse Dynamics control law

4.3 Operational Space PD control law with gravity compensation

$$\tau = g(q) + J_a^{\intercal}(q)\ K_P\ \tilde{x} - J_a^{\intercal}(q)\ K_D\ J_a(q)\ \dot{q} \tag{28}$$

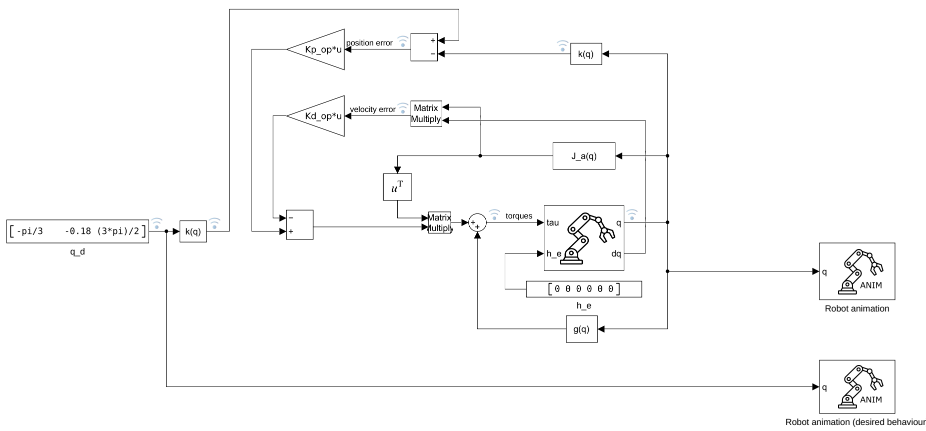


Figure 10: Operational Space PD control law with gravity compensation scheme

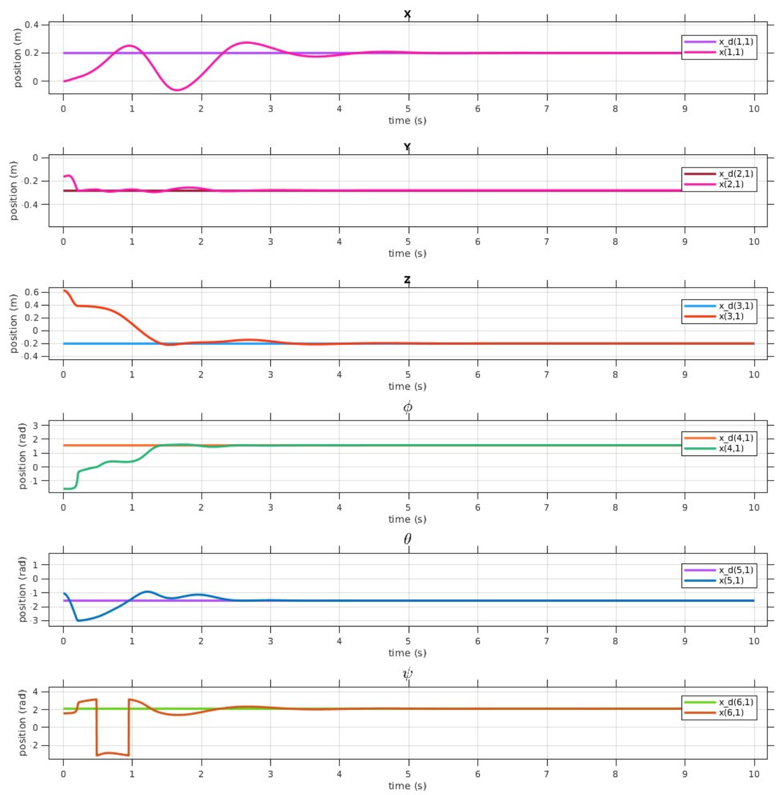


Figure 11: Operational Space PD control law with gravity compensation

4.4 Adaptive control

$$\begin{cases} \tau = Y(q, \dot{q}, \dot{q_r}, \ddot{q_r})\hat{\Theta} + K_D\sigma \\ \dot{\hat{\Theta}} = \Gamma Y^\intercal(q, \dot{q}, \dot{q_r}, \ddot{q_r})\sigma \end{cases} \tag{29}$$

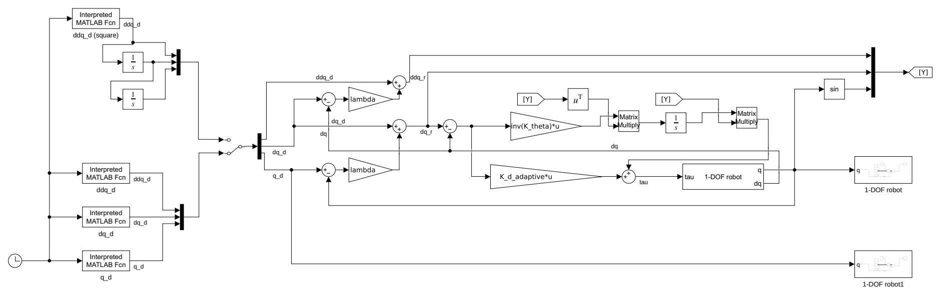
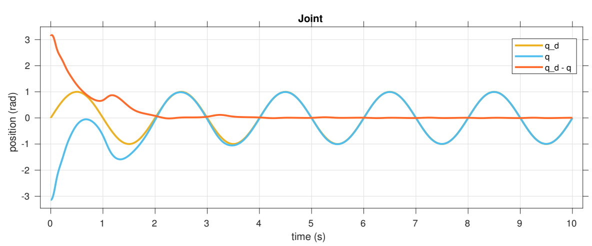
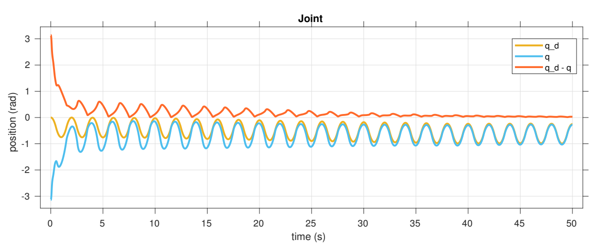


Figure 12: Adaptive control scheme



(a) Adaptive control with sinusoidal trajectory



(b) Adaptive control with square trajectory acceleration

Figure 13: Adaptive control

4.5 Active compliance control

$$\tau = g(q) + J_{A_d}^{\intercal}(q, \tilde{x})(K_P \tilde{x} - K_D J_{A_d}(q, \tilde{x})\dot{q}) \tag{30}$$

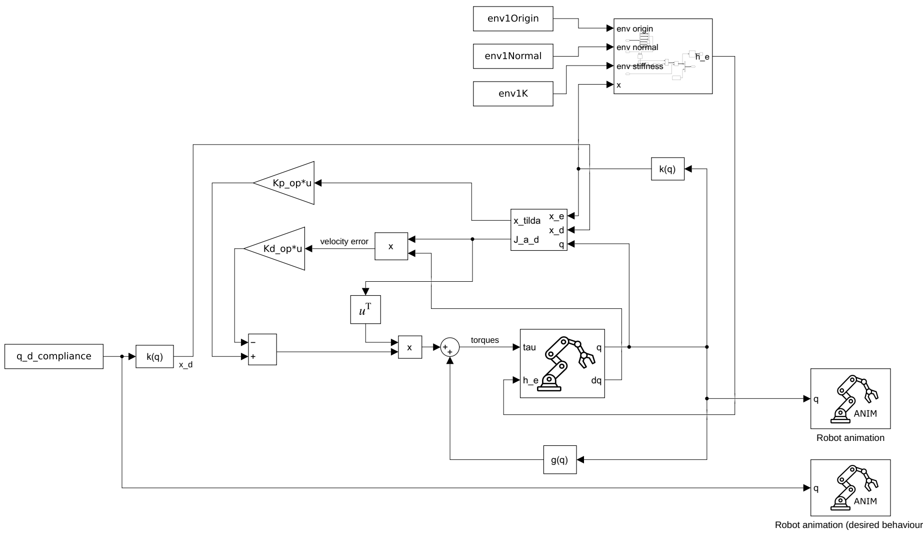


Figure 14: Active compliance control scheme

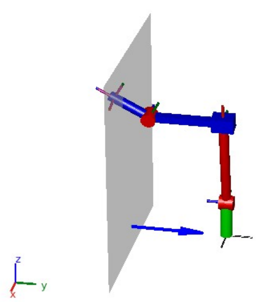
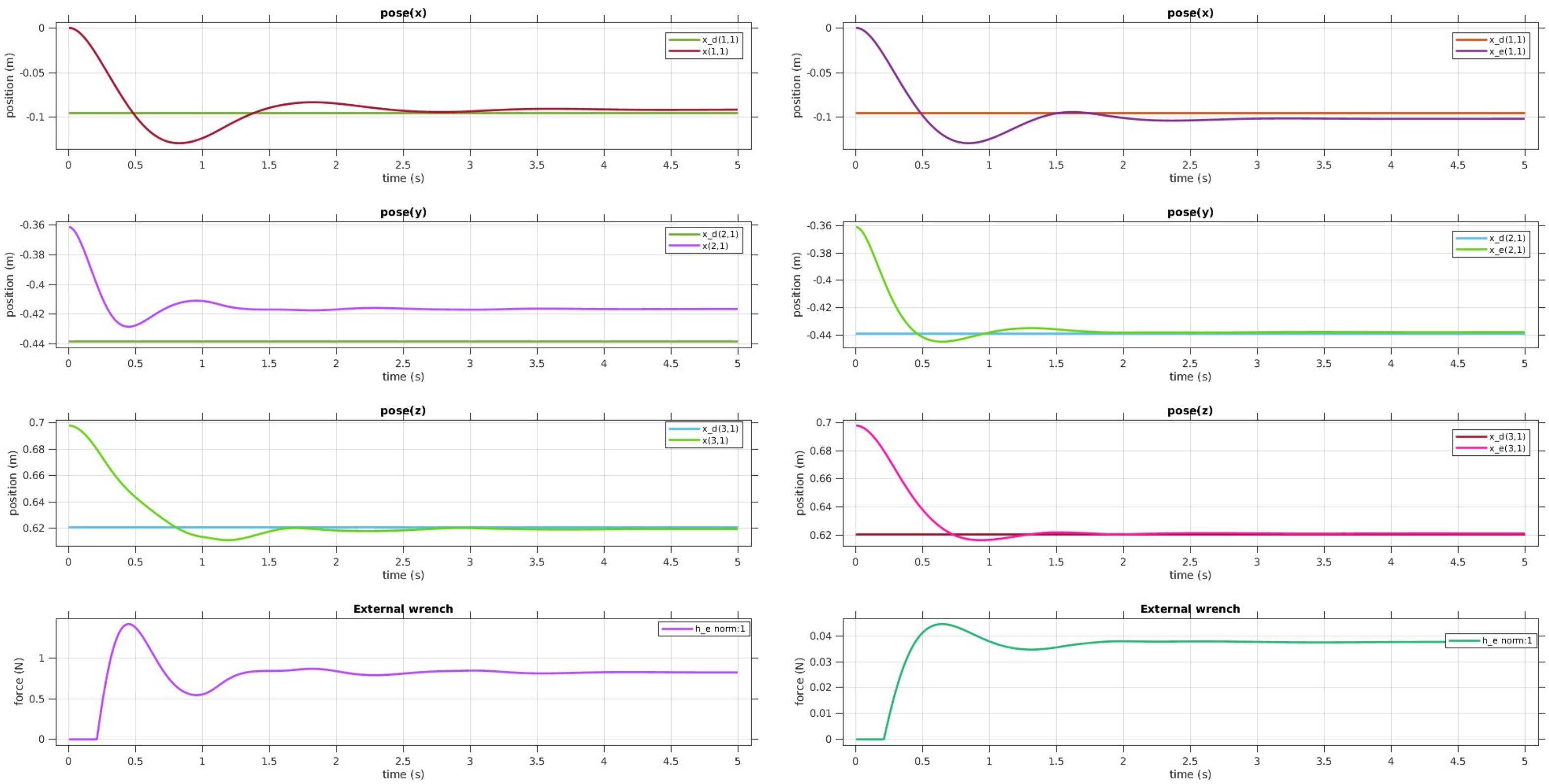


Figure 15: Environment



(a) Active compliance control
environment stiffness $K = 50\,\mathrm{N}\cdot\mathrm{m}^{-1}$

(b) Active compliance control
environment stiffness $K = 1\,\mathrm{N}\cdot\mathrm{m}^{-1}$

Figure 16: Active compliance control

4.6 Impedance control

$$\begin{cases} \tau = B(q)y + n(q,\dot{q}) + J^{\mathsf{T}}(q)h_e \\ y = J_A^{-1}M_d^{-1}(M_d\ddot{x}_d + K_D\dot{\tilde{x}} + K_P\tilde{x} - M_d\dot{J}_A(q,\dot{q})\dot{q} - h_A) \end{cases} \tag{31}$$

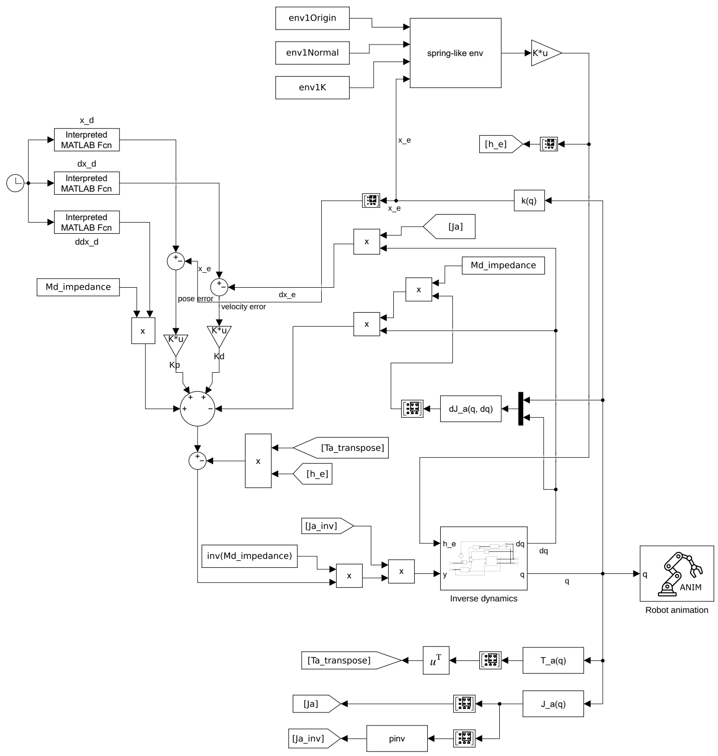
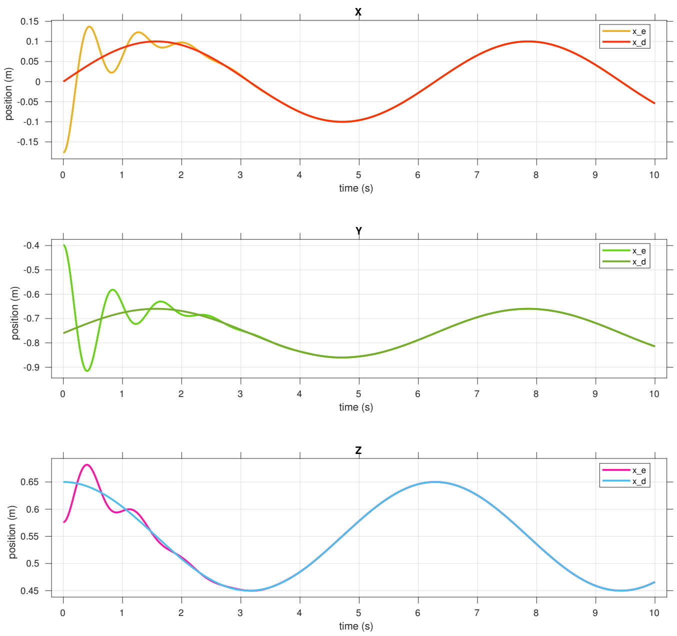
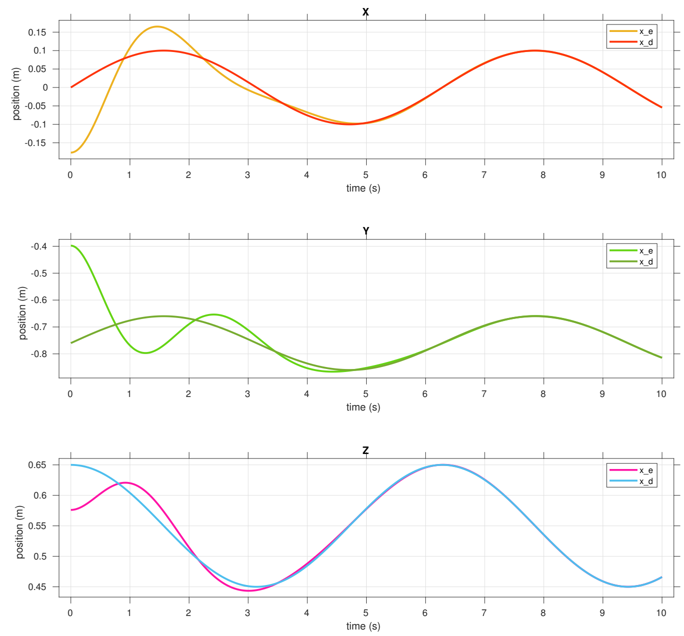


Figure 17: Impedance control scheme



(a) Higher equivalent matrices



(b) Lower equivalent matrices

Figure 18: Impedance compliance control

4.7 Admittance control

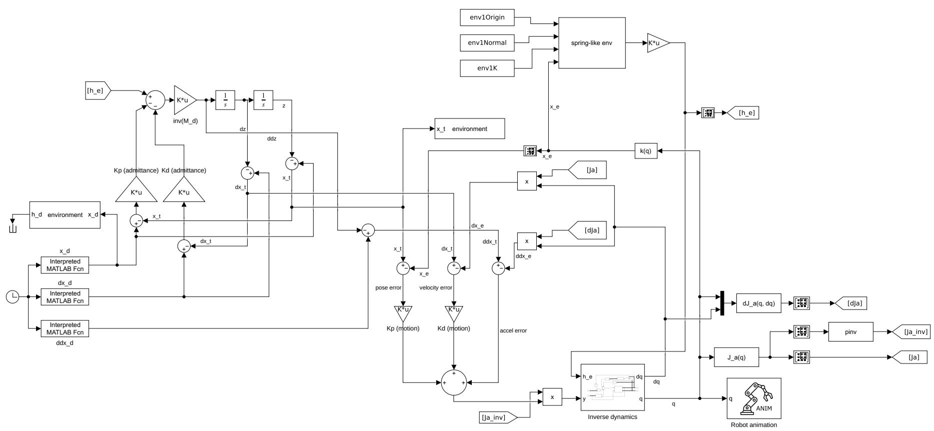
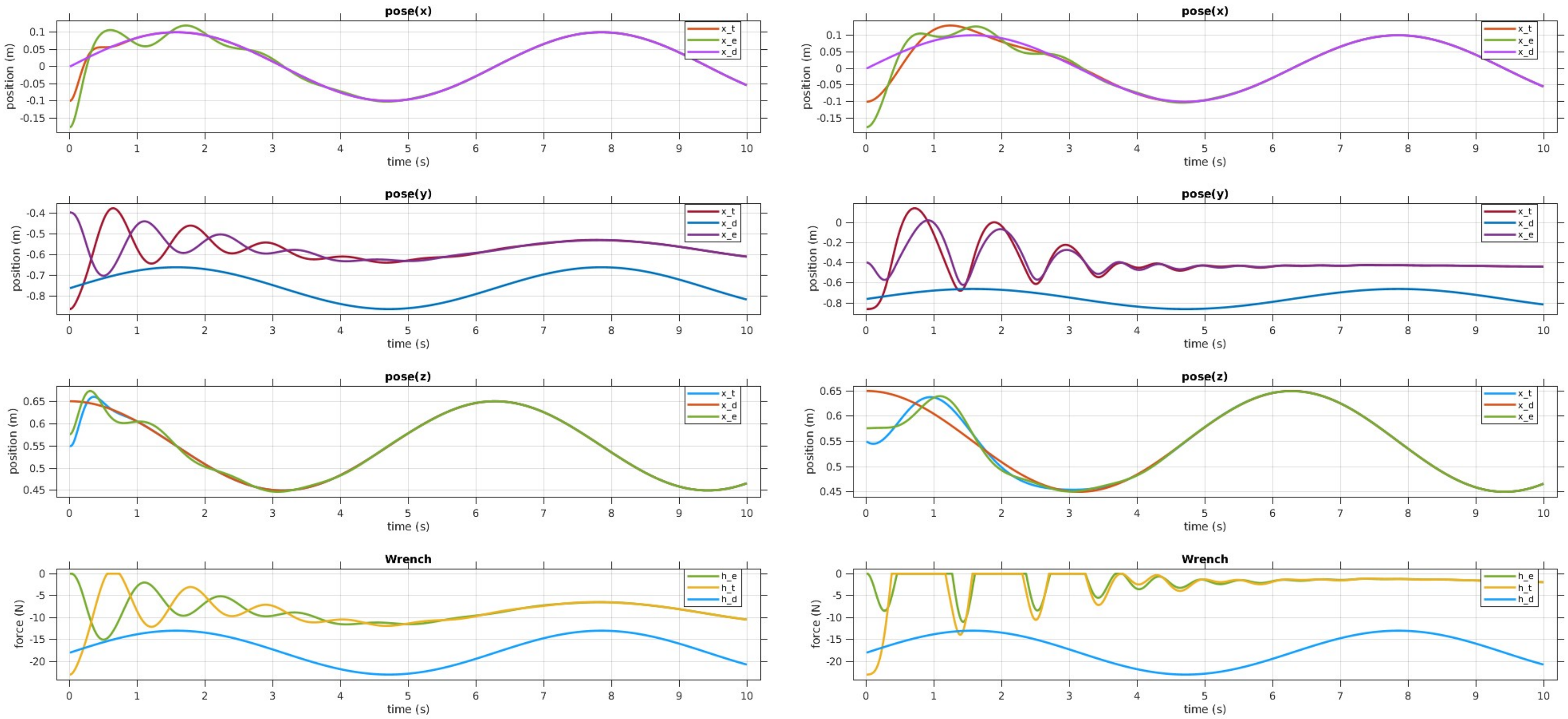


Figure 19: Admittance control scheme



(a) Stiff behaviour (b) Compliant behaviour

Figure 20: Admittance control

4.8 Force control

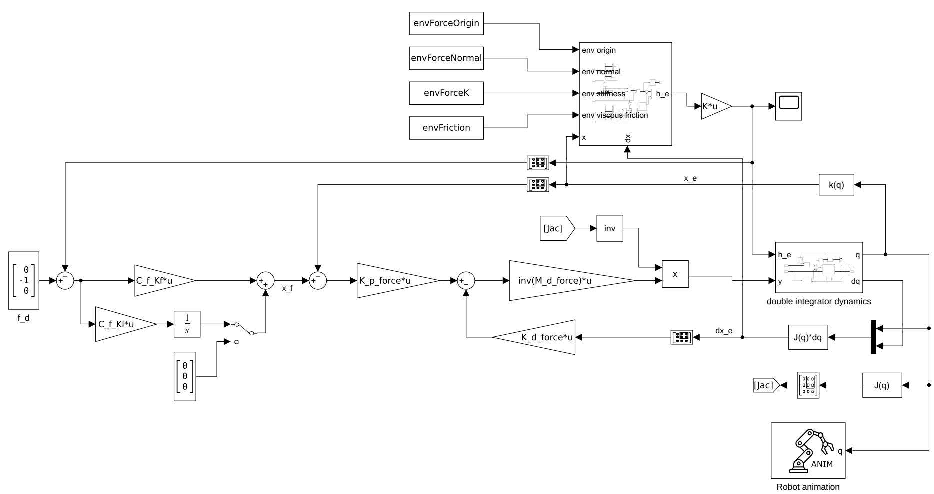
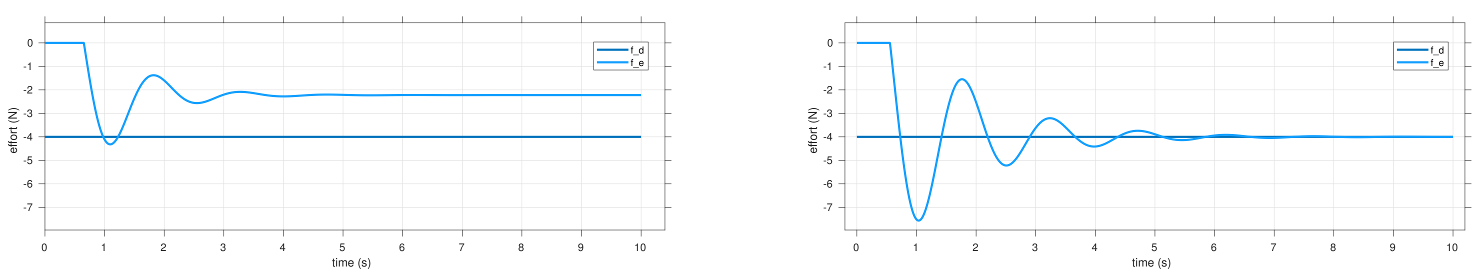


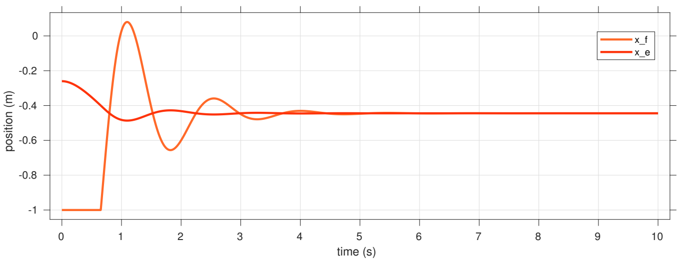
Figure 21: Force control scheme



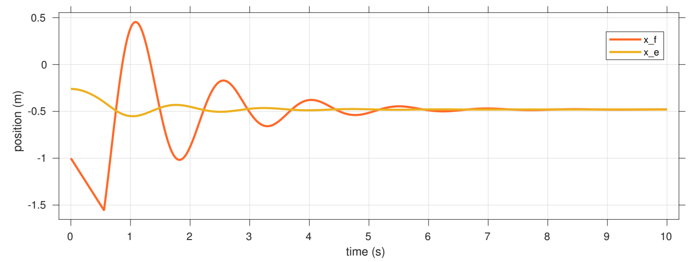
(a) $C_f(s) = 5$

(b) $C_f(s) = 5 + 3 \cdot \frac{1}{s}$

Figure 22: Force control wrench on environment normal direction



(a) $C_f(s) = 5$



(b) $C_f(s) = 5 + 3 \cdot \frac{1}{s}$

Figure 23: Force control pose (y axis)

4.9 Parallel force/position control

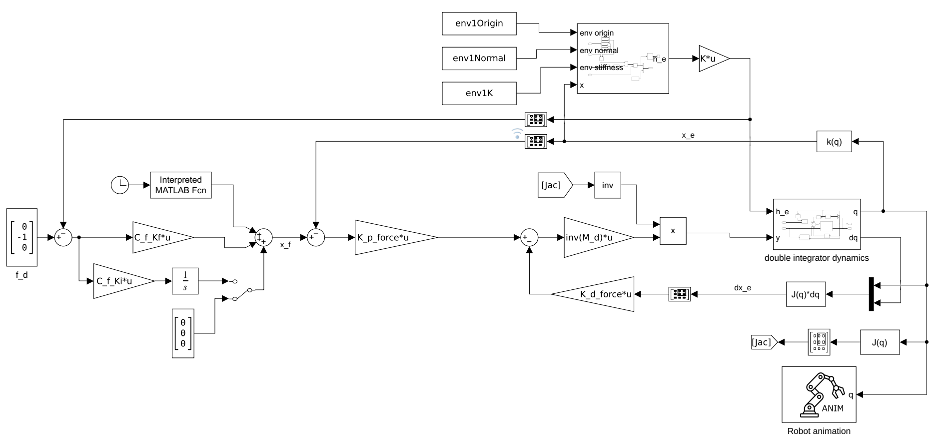
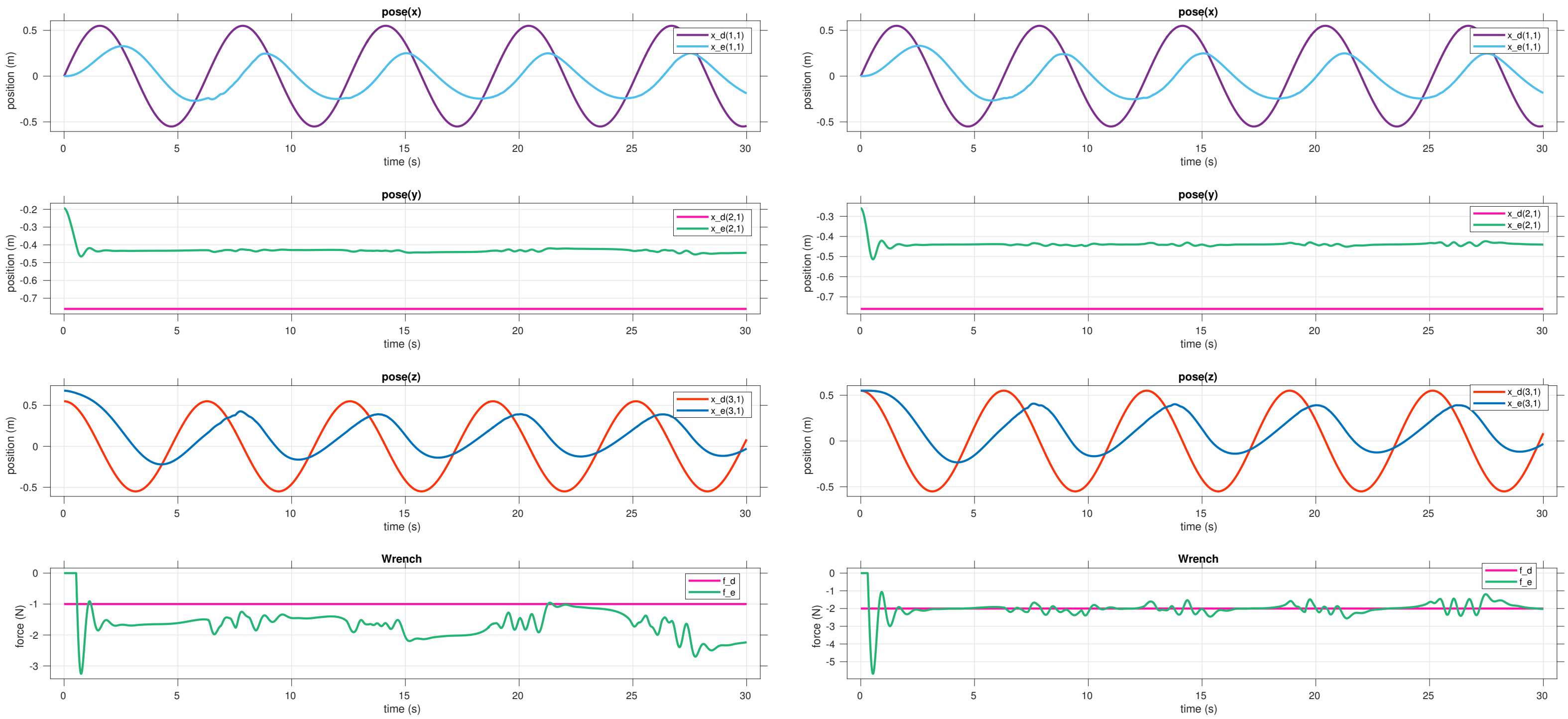


Figure 24: Force/position control scheme



(a) $C_f(s) = 5$

(b) $C_f(s) = 5 + 3 \cdot \frac{1}{s}$

Figure 25: Parallel force-position control, end-effector pose and wrench.